Plastics to Power: Producing Fuel

Report #2

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CHE 397 – Senior Design II

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Project Charter

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Introduction

Definition

Plastic is the key component of many products produced in various industries. Due to the large demand for plastic, the consumption of various plastics products has reached approximately 419 million tons in 2019. [1] The World Bank reports that the proportion of plastic waste occupied in the overall municipal solid waste all around the world has been nearly 8-12%, and the amount of plastic waste has been estimated to increase at an annual rate of 9-13% by 2025. [2]. Additionally, China has announced a ban on importing any kind of plastic waste from any other countries, which directly leads to the change of policy of plastic waste management in the US [3]. If not addresses, this becomes an issue because there will be a surplus of waste plastic that will continue to pile up in landfills and recycling facilities.

Additionally, plastic waste contributes to environmental issues and impacts the health of humans. If plastic has been broken down into small molecules, it is extremely difficult to eliminate them from the environment. Additionally, small plastic molecules negatively affect microorganisms when they accidently consume it. And these microorganisms will be consumed by many kinds of insects, especially in water. And then, the insects will be eaten by fish or livestock, and people eat them as food. Eventually, these harmful particles enter the human body.

There are several potential solutions to deal with plastic waste. A few of them are recycling, incineration, liquid fuel, plastic roads, plastic concrete, and plastic to fabric. However, each of these methods has its own limitations. For example, incineration may be the easiest way to remove plastic waste from landfills. Incineration produces electricity, but also comes with the generation of carbon dioxide.

Our solution to the rise in recycled plastic is to use that plastic to produce gasoline. Using this alternative route of plastic disposal provides some energy security by using local material to produce gasoline. This may not be the most efficient way to use recycled plastic but there are steps that are being made in order to make it a more cost effective and time efficient process.

The goal of this report is to investigate potential solutions for plastic to fuel production. This involves choosing a raw material, product, and catalyst that optimizes the process to be as efficient as possible. First, this report will provide a market analysis regarding consumed and recycled plastic as well as a market analysis on gasoline. Next, this report will provide an analysis of the process, including the catalyst. Finally, a design basis, block flow diagram, process flow diagram, and material balance will be provided.

Explanation

In this project, our goal is to design a plant that can accomplish the task of converting plastic waste into gasoline. Although plastic recycling can reduce the amount of plastic waste that is found both in the environment and in landfills, the conversion of plastic waste into fuel is still in the preliminary stages. Plastics to fuel has become a hot topic and research and the process has been attracting scientists worldwide to make the process the best that it can be.

In order to solve the problem of plastic wastes, a plan should be created to make that harmful waste into productive energy instead of having the plastic sit and waste away. As plastic is mainly a product from petroleum, the plastic can be broken down and converted into mainly gasoline fuel. This will not only solve environmental issues and health hazards, but it will help the demand for energy. Due to the growing population of Earth, there are more and more items being produced, all of which need transportation to reach consumers. For gasoline fuel specifically, the fuel can be used for vehicles, boats, generators, machinery. [4]

Building a plastic to fuel plant in the United States will be economic as the country leads the market with the highest per capita conception and collection of waste plastic, paving the way for the largest recycled plastics market. [5] This will help provide the feedstock for the plant, as the major cost is the plastic waste collection and cleaning, which can be supported by the plastic recycling facilities in the United States.

Judging Criteria

First, we need to make sure our process of removing plastic from the environment is environmentally friendly. Our original purpose is to reduce the plastic products that end up in a landfill, and further convert it into gasoline. From an overall, this idea should improve the environment because it decreases the number of plastic products in landfills.

In addition, another criterion is the process itself. We can focus on whether the process in our plant generates any pollutants, toxic gases, or if it needs a large amount of heat and energy. One source of environmental contamination that can be caused from our production process is the generation of byproducts or CO_2 gases. If it is emitted into the atmosphere without any further purifying process, it will pollute the air and will be harmful to our lungs. Large quantities of CO_2 would mean that our plant is not a zero-carbon-emissions plant. This would mean that although we are reducing the amount of plastic in the world, we are still polluting the air which gives an overall negative environmental impact. Additionally, we must identify heat and utility costs in our plant and optimize them. If the cost of utilities is greater than the total profit, this process will not be profitable.

Generally, in our project, the judging criteria are related to the environmental issues. If we can make our plant meet the above requirements, the plant will be a successive project due to its contribution to our environment.

Market Analysis

Raw Material

Plastic has become an integral part of daily life. Plastic is used for packaging food or goods, automobiles, and even furniture. However, after use, most plastic is placed in the trash and end up in landfills. Recycling was introduced in the 1960s, and by the 1980s, most major US cities had established a recycling program. During the 1990s, plastic manufacturers were required to produce polyethylene terephthalate (PET) or high-density polyethylene HDPE for easier recycling purposes. Recently, microplastics have been recognized as harmful to the environment. [3, pp. 26]. For our chosen process, HDPE plastic that has been washed and shredded will be used as the feedstock.

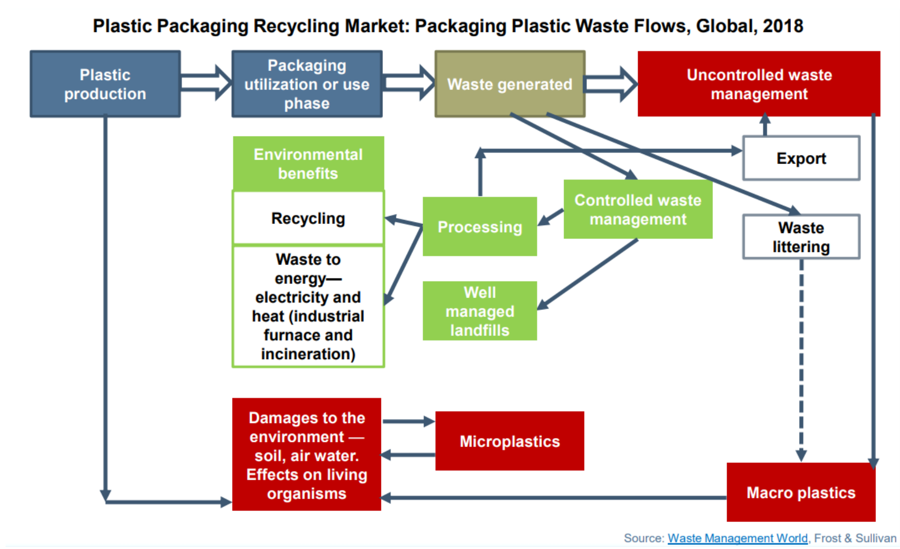
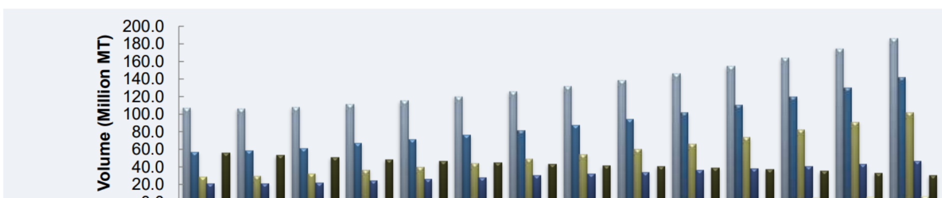


Figure 1: Process Diagram of Plastic Packaging Recycling [5, pp. 29]

According to a report from Frost and Sullivan, the amount of plastic waste generated in 2018 was 6 billion metric tons. Plastic packaging accounts for 39%, the highest percentage of plastic waste. Additionally, 79% of plastic packaging ends up in landfills. [3, pp. 28]. In the same report, Frost & Sullivan outline key global trends in plastic packaging. A few key trends are an increase in plastic water bottle usage by people in developing nations who lack access to clean water. Also, in the US, online shopping has increased the demand for packaged plastic goods, bubble wrap, and tape. Plastic packaging is also the preferred method to pack food and personal cosmetic like shampoo. [3, pp. 32]. *Figure 1* shows the process diagram for how plastic ends in a controlled waste facility, litter, or an uncontrolled waste facility.

It is also important to note that recycling plastic into new plastic is not the only method to remove plastic from landfills. Plastic can be used for roads, concrete, and textiles. Additionally, plastic can also be incinerated and converted into electricity or directly converted into hydrocarbons used for petroleum fuel [3, pp. 153]. The global demand for plastic is also increasing. *Figure 2,* below, shows the increasing demand for plastic globally. In 2025, the global production of plastic is estimated to be 139.3 million tons [3, pp. 14].



	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Generated	107.5	106.3	108.2	111.8	115.8	120.6	126.0	132.3	139.3	146.8	155.4	164.8	175.3	186.8
Collected	57.4	58.8	61.8	67.3	71.5	76.6	81.9	88.1	94.7	102.2	110.9	120.5	130.9	142.4
Recycled	28.8	30.4	33.0	37.1	40.7	44.8	49.4	54.5	60.2	66.8	74.3	82.6	91.9	102.1
Incineration & WtE	21.2	21.5	22.7	24.8	26.6	28.6	30.7	32.8	34.7	36.7	38.9	41.2	43.9	46.9
Waste to Fuel	0.5	0.8	1.0	1.3	1.7	2.1	2.5	3.0	3.5	4.1	4.7	5.4	6.2	7.0
Landfill*	57.0	53.7	51.5	48.6	46.8	45.1	43.4	41.9	40.9	39.3	37.5	35.6	33.3	30.7

Figure 2: Recycled Plastic Global Volume Forecast [3, Pp.14)

With an increasing demand for plastic, Frost & Sullivan have also identified key areas to improve plastic packaging recycling. For plastic manufacturers, it is important the plastic is labeled with their correct composition number for easier and accurate disposal. Innovations in technology such as Near Infra-Red (NIR), X-Ray Diffraction (XRD), and X-Ray Fluorescence (XRF) and smart recycling bins and vending machines could improve sorting. In addition, scientific innovation in compatibilizers, which are used to blend incompatible resins, catalysts for chemical recycling, which is currently a very expensive process could also increase recycled plastics. Lastly, Frost & Sullivan highlight that the process of turning plastic into fuel process is in the early stages of development and could lead to an additional avenue to remove plastic from landfills. Two companies, Quantafuel, and Polycycl, currently process plastic to fuel. [3, pp. 34, 172].

High density polyethylene (HDPE) is labeled with the recycling symbol 2. It is derived from petroleum and used for products such as milk jugs and shampoo bottles [6]. *Figure* 3, below, shows the demand for HDPE. In the Frost & Sullivan report, for 2018, 76.9% of HDPE is collected in the Americas and 33.4% is recycled [3, pp. 44]. In the Americas in 2018, 7.49 million tons of HDPE was generated, 5.8 million tons was collected, and 2.5 was recycled [3, pp. 109]. The EPA reports that in 2016, 1,270 thousand tons of HDPE was recycled in the United States [7].

Plastic Packaging Recycling Market: Collection Rate and Recycling Rate—HDPE, Global, 2018

Year		Asia-Pacific	Europe	Americas	MEA
2018	Collection Rate	44.2%	81.0%	76.9%	38.6%
	Recycling Rate	24.6%	50.7%	33.4%	17.7%
2030	Collection Rate	72.8%	92.0%	95.4%	62.2%
	Recycling Rate	53.7%	77.5%	58.2%	41.1%

- The collection rate of HDPE plastic packaging is the highest in Europe which is about 81.0% followed by Americas 76.9%, APAC at 44.2% and MEA 38.6% respectively. MEA exhibits lower collection rate due to lack of collection infrastructure in Africa and low awareness in Middle-East.
- The recycling rate is the highest in Europe which is 50.7% while MEA is the lowest with a recycling rate of 17.7%.

Figure 3: Global Polypropylene Recycling Rates [3, pp. 44]

Products

The main product produced in this process is unfinished gasoline. After production, unfinished gasoline is sold to blending facilities which blend in ethanol and other additives. After the blending process, gasoline is considered finished, and can be sold to consumers. Other by-products that are produced during catalytic cracking are C_1-C_4 hydrocarbons, CO_2, and char.

Gasoline is the primary fuel used for transportation in United States. In 2018, the US consumed 143 billion gallons of motor gasoline and 186 million gallons of aviation gasoline [8]. The majority of gasoline sold in the United States is a blend containing 10% ethanol by volume [8].

Gasoline, and other liquid fuels, are derived from crude oil in a refinery. Refineries produce a product called unfinished gasoline or gasoline blendstock. Unfinished gasoline on its own is unsuitable to be used in an engine. It must be blended with other additives to become finished motor gasoline. In the United States, most of the blending occurs at blending terminals, not at refineries [9]. Most commonly, a 10% fuel ethanol is blended with gasoline to meet the Renewable Fuel Standard. The goal is to reduce greenhouse gas emission and provide energy security. Certain retailers will add detergents to gasoline to create their own formulation [9].

Gasoline is sold in three types of grades, regular, midgrade, and premium. Different companies may have different names, but each grade represents a different octane number. Each fuel is tested to determine the pressure when the fuel will auto-ignite. The octane rating is based on the auto-ignite pressure. The octane number seen at the pump in a yellow square is an average of two tests: motor octane rating (MOR) and research octane rating (ROR). Generally, regular gasoline is 87, midgrade is 89-90, and premium is 91-94. [10]

Gasoline prices vary over different states due to a variety of factors. These factors include distance from supply, disruptions in supply, and competition from other retailers. Additionally, each state has their own environmental regulations regarding reformulation, storage, and transportation. For example, some states have requirements that the fuel contain additives to reduce smog. The Energy Information Administration reports that one-third of fuel of gasoline sold in the United States is reformulated [11]. These varying regulations and supply factors cause prices to vary.

The Energy Information Administration reports the price of gasoline for different regions in the United States. See *Figure 4,* below, for map of each district. In 2019, Energy Information Administration reports that the West Coast district has the highest average annual price at $3.337 per gallon of gasoline. The lowest average annual price per gallon is the Gulf Coast district ($2.283 per gallon) [11].

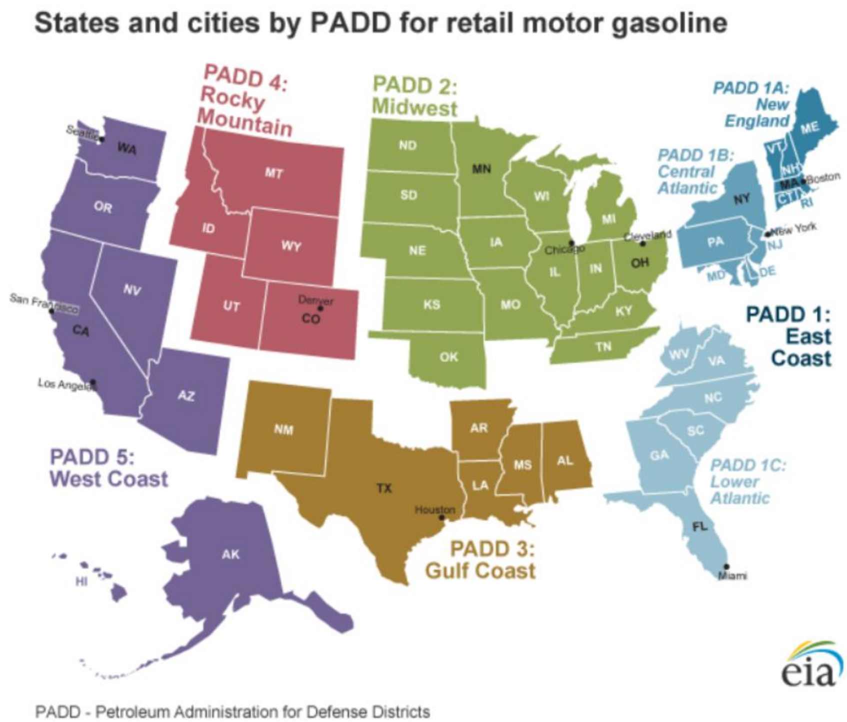


Figure 4: EIA Map of PADD Districts [11]

Another by-product is C_1- C_4 light hydrocarbons. These hydrocarbons have a few possible uses. First, the hydrocarbons can be burned and used to generate heat for processes inside the plant. Or, the hydrocarbons could be purified and sold. Both options will require investment. If the hydrocarbons were burned, the stream would need to be cleaned to make sure emissions are compliant with regulations. If the hydrocarbons were sold, equipment for separating and purification would need to be purchased.

The remaining by-products are CO_2 and char. CO_2 is released during the pyrolysis process. The CO_2 will need to be captured. The char can be further used for an additional complete combustion process, or it can disposed as a waste product.

Value Chain

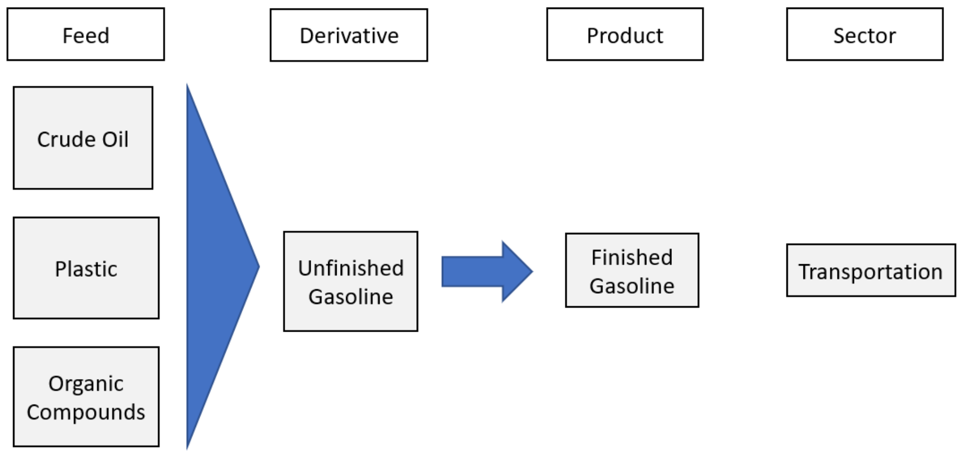


Figure 5: Value Chain

Figure 5 shows the value chain for the competing processes. Competitors of this process are refineries who produce gasoline from petroleum.

Process Description

Competing Plastic Recycling Processes

Overall, there are six different plastic waste management options that are well known. They are recycling, incineration, liquid fuel, plastic roads, plastic concrete, and plastic to fabric. Recycling involves sorting, transporting, and converting plastic into raw materials. Incineration converts the plastic waste into energy and heat. Liquid fuel is formed through chemical synthesis as plastic waste as the feed. Plastic roads are made by incorporating plastic bags and bottles into the asphalt to increase durability. Similarly, plastic can also be incorporated in concrete for increased durability. Plastic, specifically polyethylene bottles, can also be processed into polyester yarn for clothing.

Before touching on the processes described above, one might ask why plastic isn't recycled to produce more plastic? This is because every time that a plastic is recycled, the quality of the plastic decreases. This is due to the polymer chain becoming shorter and shorter each time that the material is broken down [12]. To make recycled plastic stronger, a new polymer is mixed in. However, even with addition of the new polymer, plastic can only be recycled two to three times before the quality of the material makes it no longer usable.

One of the benefits of recycling plastic is that recycled plastic uses one-third of the energy than if the plastic was made from virgin materials [13]. However, currently, it is more expensive to produce recycled plastic than it is to make new plastic. It was estimated that recycled plastics cost $72 more per metric ton than new plastic [14]. Additionally, petrochemicals are currently abundant, making it cheaper to make new plastic. [14] Recycling plastic has increased due to the large push to include recycled plastics in other materials or objects.

In recycling, there are four major steps. First, the plastic waste must be collected from residents. Then, the collected waste is transported to the recycling facility. Next, the plastic waste is sorted by series of hand separations and machine separations. Finally, the plastic is either mechanically or chemically, converted into pellets to be used in future plastic. [5]

Plastic roads are a new developmental technique in recycling waste plastics such as bags, packaging material, bottles, and cups. These plastics are shredded and mixed with asphalt for the construction of roads. One company, PlasticRoad, is optimizing the load bearing capacity of the plastic and asphalt material to make plastic roads become a reality. Their design is hollow underneath the surface of the road to store drainage water which prevents flooding. [5] Furthermore, plastic roads produce lower carbon emissions and have a longer lifespan, about three times than the traditional road. The longer lifespan lowers frequency of remodeling roads. Additionally, the lightweight plastic makes the construction of the road easier and faster. However, currently, plastic roads are still in their pilot stages. The first application is of a 30-meter road for bicycles. Further research must be done to design roads for heavier loads such as cars and trucks. Additionally, further research is needed to determine how sound vibrations effect driving on plastic roads. For cold climates, analysis during winter will need to be conducted regarding traction. [15]

An alternative way to re-use plastic is clothing. Clothing made from recycled plastic is a very common process. Polyester from plastic bottles are synthesized into plastic fibers. The fibers can be woven to make pants, jackets, and hats. For example, by using a plastic waste feed, Patagonia, a clothing company, has reduced their raw material of oil by 20,000 barrels, reducing their dependence on petroleum. [5] However, after the clothing is worn, if it is not sent back to the company to be recycled again, it is sent to the landfill. Since the clothing is not biodegradable, this method of recycling is not solving the issue of growing plastic waste.

Furthermore, when washing polyester clothing, microplastics are deposited into the water supply. Then, fish consume the microplastic, and ultimately, microplastics are found in humans. [16]

Another way to re-use plastic is to produce liquid fuel, petrochemicals, and lubricants by chemical synthesis. One example of liquid fuel production from industry is company called Recycling Technologies. They have created a machine, RT7000, that takes all plastic waste, including typical unrecyclable plastic, to produce an oil called Plaxx®. Their goal is to replace feedstock from fossil fuels with Plaxx® derived feedstock for new plastic and wax production. [5] Plaxx® can also be used as fuel for marine vehicles and for engines and burners. However, there is only at 75% yield of Plaxx® from the RT7000. [17]

Chosen Plastic Recycling Process

The chosen process for plastic waste management will be the liquid fuel route, where the product will be a synthesized hydrocarbon fuel mixture. The chosen process will be similar to the process that is being performed by the Recycling Technologies. Overall composition differences in the end product will depend on the raw material, reactor and a possible catalyst. The other five possible processes had critical flaws in which the end product would end up in landfill eventually or even cause to much pollution. Recycling can only be done so many times on a plastic before it degrades. The process of turning plastic into clothing is not efficient because the plastic will eventually end up in a landfill. Plastic roads and concrete are still in the early phases of development, and the incineration process causes too much pollution. On the other hand, plastic waste to fuel offers a product that is in large demand and will sell for great profits assuming that the process is economical. Another benefit of plastic conversion to fuel is that it focuses on pyrolysis, thus the only emissions will come from heating the reactor. Pyrolysis is where high temperatures are used in an inert atmosphere to proceed the reaction. Catalytic pyrolysis shows higher conversion rate of the feed to product, so a catalyst for the process is favored. [18]

Competing & Chosen Catalyst

Currently in industry, zeolite catalysts are very common because they do a good job of reducing the temperature needed for pyrolysis. However, there are several drawbacks when using zeolites. Almost all potential catalysts for this process have drawbacks. Overall, the best catalyst for this process is a catalyst that reduces the amount of energy needed for pyrolysis and can speed up the overall process. Other minor factors to consider are the temperature that the catalyst operates at and the liquid recovery. There were three different catalysts that were contenders for the pyrolysis process: silica–alumina, ZSM-5, and MCM-41. The catalyst is added to the liquidized plastic to increase the process efficiency, quality of the oil, and to reduce the time and temperature of the reaction.

One of the most common catalysts used for the pyrolysis of plastic is the ZSM-5 zeolitic catalyst. ZSM-5 can be found both on its own in the environment and can also be made synthetically. One of the benefits of making the catalyst is that it can be made to have a specific mesoporous structure. Changing the mesoporous structure of the catalyst ad increasing or decreasing the pores will lead to recovery of different hydrocarbons. It is also important to note the difference between natural and synthetic zeolites. Natural zeolites tend to give a larger liquid yield than synthetic zeolites. The natural zeolites produce a liquid yield of 54% with a very large char yield of 33.2%. Synthetic zeolites, HZSM-5, on the other hand gives a liquid yield of 50% with a smaller char percentage of 27.2% [19]. This means that a synthetic zeolite gives a higher gas yield and produces a lower percentage of char. Since liquid fuel is the main product, it would be best to use a natural zeolite. Natural zeolites also tend to have a lower acidity, and microporous structure. By increasing the Brunauer-Emmett-Teller (BET) surface area, the highest reported liquid production using zeolites was reported at around 80% [19], but the process involved the use of both synthetic and natural zeolites in a multi-step process. Using both natural and synthetic zeolites would increase the cost of the process making it less economically friendly and less efficient.

In order to increase the yield of gas, another option is to impregnate the catalyst with a metal. Impregnating the catalyst with a metal would cause the surface area to increase. The feed material can be various types of plastics rather than only polystyrene with the ZSM-5 catalyst. Metal based zeolites, when used with high density polyethylene, produce little to no char. This is beneficial because it limits frequency of stopping the production process to clean the char in the reactor. Since gas is not the desired product, metal zeolite is not a good solution.

The other well-known catalyst that is used for the pyrolysis of recycled plastics is a $SiO_2 - Al_2O_3$ catalyst. This catalyst is known for working well with Polypropylene as the feed material. $SiO_2 - Al_2O_3$ when used with polypropylene had a conversion of 25.8%. During this process, liquid fuel is 60.3% of the product stream. [20]. The rest of the product is gas. Because of the low recovery of liquid, this catalyst should be considered with a different feed material. When using a feed material of HDPE, the amount of gas produced from the reaction increased to 59.63 %. [21] This has to do with the porosity of the catalyst. If the pore size is too small, a bulkier molecule like polypropylene will not fit in the catalyst and the reaction will only be partial or nonexistent.

Silica-alumina catalyst has a high acidity which increases gas recovery. As more catalyst is added, the acidity increases which leads to degradation of the primary pyrolysis products. This causes the pyrolysis molecules to become even smaller and break down further into smaller hydrocarbons. [21] This then changes the composition of the product. Due to liquid product being the desired product of the reaction, silica-alumina was not chosen.

The final, most common catalyst for the pyrolysis of plastics is MCM-41. MCM-41 is a low-density silicate catalyst and it can be used as a surfactant [22]. The product can be bought in different diameters, which means that the surface area will vary. The diameter size affects how much gas or liquid product is recovered. Using MCM-41 is essential when looking for a product that contains heavy hydrocarbons. Zeolites limit the hydrocarbons that can be found in the product, which is why MCM-41 is used as an alternative when producing heavier hydrocarbons. [20] When heavier hydrocarbons are desired, a mix of MCM-41 and ZSM-5 are used. This then also limits the use of the feed material that can be used. When mixing the two catalysts, high-density polyethylene is desired in order to produce a quality, liquid product with high yield.

Just like with the zeolite, there is also the option of impregnating the MCM-41 catalyst with a metal. In the next described case, MCM-41 with aluminum will be discussed. When comparing Al-MCM-41 and HZSM-5, a synthetic zeolite catalyst, HZSM-5 has a higher conversion of polyethylene to product. This is due to the high acidity of the Al-MCM-41 catalyst. As described above, a higher acidity and higher surface area will lead to a lower liquid yield. However, when considering polypropylene as the starting material, Al-MCM-41 achieved a conversion of 99.2% while HZSM-5 only had a conversion of 11.3% [19]. This occurs because polypropylene is bulky and cannot fit into the small pores of the HZSM-5 catalyst. Polyethylene would be a better fit for the HZSM-5 catalyst. This point emphasizes how important knowing the pore size of the catalyst is. A smaller pore catalyst should use a feed material that has smaller molecules, like polyethylene.

MCM-41 catalyst is a great option and has made a lot of progress in the plastic to fuel industry, and is one of the front running catalysts. This is the reason why it was chosen for our process. MCM-41 not only has a high conversion rate, but also produces heavier hydrocarbons that fit into the diesel and gasoline categories.

Now that MCM-41 has been chosen, it is important to consider the feed material that will be used. This should be considered not only from a reactivity side but also from the angle of being able to collect the amount of plastic desired. Polyethylene is more abundant than polypropylene. In fact, 97% of bottles produced in the US are made from PET and HDPE while only 1.9% of the bottles are made from polypropylene. This means that the chosen catalyst needs to be compatible with a plastic that is in high demand, like high density polyethylene. 98.8% of recycled plastic is comprised of PET and HDPE while on 1.1% of recycled plastics are made from PP. [23] It is also important to recognize that polypropylene has a higher boiling point than polyethylene. Polyethylene has a boiling point of 120°C [24] while polypropylene has a boiling point of 171°C [25]. This means that a higher temperature will be needed on the reactor. This will lead to more energy for the pyrolysis reactor but also on the cooling water/refrigerant needed to cool the liquid or compress the gas after pyrolysis.

Competing & Chosen Reactor

The type of reactor was chosen based on system requirements for a quick, semi-batch process. The reactor needs to equally distribute heat around and allow for an inert gas atmosphere. The types of reactors considered are fluidized bed reactors, large batch reactors, and fixed bed reactors. Fluidized reactors offer a good semi-continuous process and allow for excellent constant mixing of catalyst and feed. Batch reactors can process large amounts of raw material, however because of the speed of the reaction, the batch process would be less favored. Finally, fixed bed reactors also offer excellent heating of the mixture and aid in a semi-continuous process. For the catalytic cracking process, higher interaction with the catalyst and raw material would offer better selectivity [26]. The best reactor for this process is a conical shaped fluidized bed reactor because it offers large heat transfer between phases, scalability, and effectiveness for petroleum-based reactions [26].

Overview

The overall process can be seen in the block flow diagram in *Figure 6*. Initially, the HDPE raw material will be received by truck loads and placed in storage. From there, the plastic will be shredded, washed, and then dried. The prepared HDPE will be mixed with the MCM-41 catalyst pellets and added to the fluidized bed reactor at set intervals. A gas stream containing C_1 – C_9 carbon chain molecules and a second liquid/residue stream would leave from the reactor. The liquid/residue stream would pass through a filter to catch any spent catalyst. The spent catalyst is sent to a regenerator and mixed back into the HDPE feed. The gas stream and filtered liquid residue streams will enter a distillation column for further purification. Both streams will be fed in at different stages and the products of distillation are propane and butane, gasoline and liquid mixture known as 'pitch'. Ideally, the three product streams are fully separated out and stored in large units. Gasoline will be sold for profit, whereas the propane, butane and pitch will be used for heating in the system.

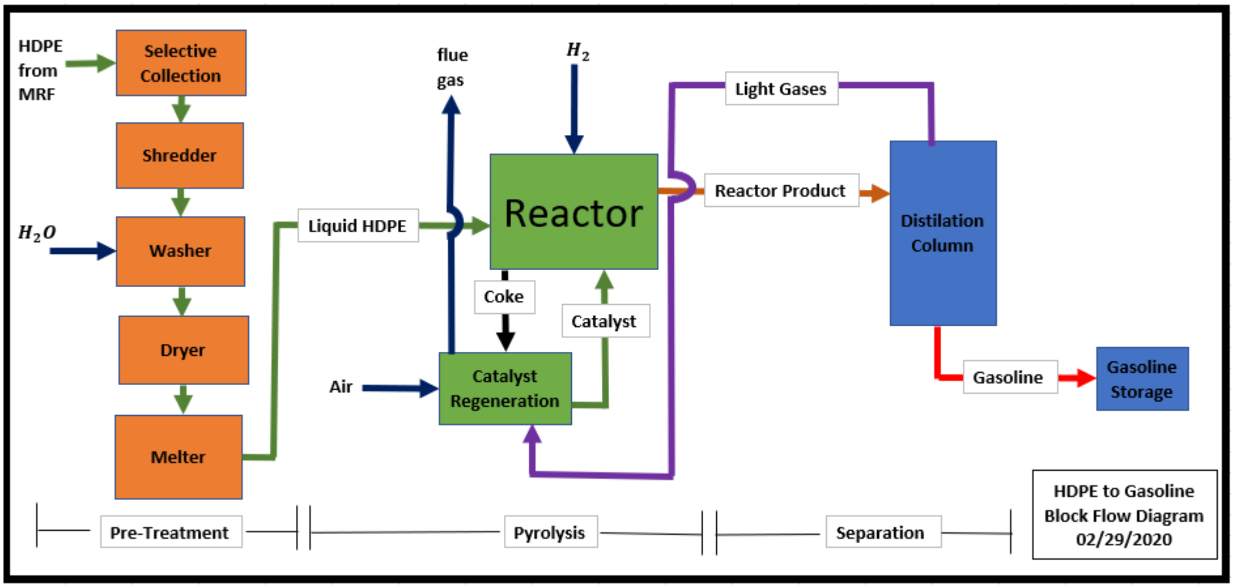


Figure 6: Block Flow Diagram

Process Assumptions

Some assumptions were required for this process because some information is proprietary, as this is a new technology. Experimental data and research were used to justify system assumptions. The first assumption is that the process will have a 100% purity HDPE raw material input, minus any impurities that will be washed away. Generally, there would be some error in the sorter process that might lead to other plastics in the feed. However, 100% pure HDPE raw material is assumed because the percent error would be small, dynamic, and hard to measure. Another assumption is that the reaction is scalable with what the results showed in researched experimental data. For the purpose of the catalyst regeneration, it was assumed that hot air will burn off the coke into mainly CO_2 with some NO_x. Any heaters in the system will be heated with a medium pressure steam, assuming it comes in at 212°C under a pressure of 20 bar [27]. Cooling water will be utilized as the cold utility for the system, assuming it comes in at 30°C [27].

Operating Conditions

The raw material selected for this process is high density polyethylene (HDPE). This feed will be initially delivered to the facility by regularly scheduled truck loads from a local recycling facility. The goal will be to process a total of 50 metric tons of HDPE in one day, where deliveries will be made every two weeks by truckload in bale form. Upon arrival, the recycled HDPE bales will be placed in a large storage area before it is further processed into the desired products. Having an initial receiving storage unit will help prevent raw material shortages or any unplanned for events at the facility that may call for a shutdown.

The HDPE may have been cleaned already at the recycling facility; however, in order to avoid contamination, the plastic will be thoroughly washed. Before washing the plastic, it will be shredded into small spherical particles of about 5mm in size. Small particle size will increase reactivity with the catalyst during the pyrolysis step of the process [18]. Then the plastic will be washed. The wastewater will have to be treated according to local state standards. In the final step of preparation, the plastic will be dried as to not introduce any moisture into the pyrolysis step.

Once the HDPE has undergone the necessary preparation, the next step in the process is pyrolysis. The plastic will be fed into a conical shaped fluidized bed reactor where it will be mixed with a mesoporous material (MCM-41) catalyst in order to increase selectivity for gasoline. The chemistry in the reaction is simple. Long polyethylene carbon chains are split or 'cracked' into smaller molecules. This can occur at either end of the polymer or at random points in the chain. A basic reaction mechanism pictured in *Figure 7*, from Appendix 2, is below. [18]

$$M_n^* \rightarrow M_{n-1}^* + M$$

$$M_{n-1}^* \rightarrow M_{n-2}^* + M$$

$$M_n^* \rightarrow M_x + M_y$$

Figure 7: Cracking Mechanisms of HDPE Polymers [18]

The first reaction in *Figure 7* features a long polymer chain cracking into two molecules. The second reaction demonstrates how the process will continue breaking down the original polymer chain. Alternatively, as depicted in the third reaction, the polymer chain can be split anywhere giving two separate, but shorter polymer chains. This is how you turn a long polymer such as HDPE into short C_1 – C_9 fuel compounds. When these polymer chains undergo cracking, they leave an open reactive site. This can lead to reverse reactions occurring where the chain reconnects instead of being further cracked. To prevent this a hydrogen utility stream will be added to the reactor, so that the carbon chains will instead react with hydrogen and not each other.

For optimal selectivity to occur under the selected conditions, the catalyst and HDPE will be mixed in a 1:100 ratio, respectively [26]. The pyrolysis step in the process will run at 360°C and at atmospheric pressure. In order to prevent the presence of oxygen in the stream, the pyrolysis step will be completed in an inert atmosphere. To accomplish this, nitrogen will be pumped into the reactor for the duration of the reaction. Under these conditions, it is expected that product will have the selectivity as featured in *Table 1*.

Table 1. Selectivity of Pyrolysis Step Products [28]

Product	wt% of Feed
Gases	23.4%
Gasoline	61.4%
Liquid	6.1%
Residue	9.1%

The 23.4% ‘gases’ product is composed of C_1 – C_4 carbon chain molecules, and the 61.4% gasoline product is composed of C_5 – C_9 carbon chain molecules. Both products exist in a complete vapor state, which makes later separation easier.

The catalyst utilized in the pyrolysis step is MCM-41, an aluminosilicate originally developed by the Mobil Oil Corporation [29]. MCM-41 is commonly used in catalytic cracking and separation processes and it is known for its cylindrical type shape [29]. For the purpose of pyrolysis, the catalyst size will be about 125 – 180 micrometers in particle diameter. Spent catalyst from the reactor will be caught by a filter when leaving as part of the liquid phase pyrolysis product and will then be sent to a regenerator. The amount of coke produced is 2.2 weight percent of the raw material feed to the pyrolysis reactor, which will most likely be taken out in the same filtration step as the catalyst. The regenerator will continuously move the catalyst and coke mixture under a hot air blower [30]. The catalyst will then be cooled by cooling water because the newly regenerated catalyst will be very hot [30]. Then, the regenerated catalyst will be added back in correct proportions to the raw materials.

The pyrolysis desired product is mostly gas and the residue and liquid output are unwanted products. After pyrolysis, the streams containing liquid and gas will undergo fractional distillation in a distillation column. The gas mixture will enter the middle of the distillation column and the liquid mixture will enter the bottom stage of the column at ideally the same temperature (360°C) from which it was at in pyrolysis. The column's temperatures will range from the 360°C feed to just below 35°C where gasoline will have completely condensed. The vapor fed into the column will travel up through the column trays and condense based on each hydrocarbon's boiling point temperatures. The column will have three output streams [18]; the bottoms will consist of a liquid 'Pitch' mixture, gasoline will leave from the middle of the column, and finally a gaseous mixture of propane and butane will leave from the top. Gasoline, which has a lower boiling point than the feed, will travel up the column along with the short carbon chain gases. Under the atmospheric conditions of the column, gasoline will mostly condense out between temperatures of 200°C – 35°C. Propane and butane will not condense and exit as the top product since their boiling points are well outside of the column's operating temperature. A partial condenser system will be added immediately at the output of the top product, which will feed into a reflux drum. The reflux drum will allow for gasses to separate out from any liquefied material which is going to be fed back into the top of the separation column.

The gasoline mixture from the column will be sent through a condenser to lower its temperature for large scale storage. Large tanks will contain the gasoline until it is sold. The bottoms pitch product, propane and butane from the column will be also cooled through a heat exchanger where it will then be stored on a large scale.

The fluidized bed reactor will require a large amount of heat input to keep the temperature constant at 360°C for catalytic cracking to occur. According to selectivity experimental data, used as a basis for the process, of the 23.4% feed weight gas product, 16.7% is butane and the remainder of the gas is propane [28]. The propane-butane and pitch products will be used as fuel source for the pyrolysis and dryer units; however, this will produce CO_2 which will need to be regulated as per local standards.

Appendix 1 - Design Basis

The size of a plant is one of the most important design specifications because it determines the production, equipment size, and location. The sizing of the plant is based on the size of Quantafuel's existing pilot plant in Denmark is 60 metric tons of recyclables per day [3, pp.164]. We have decided to design a pilot plant that will use 50 metric tons of recyclables. Also, it must be noted that the raw material plastic must be HDPE (#2 recycling number only) The process should yield 61.4% gasoline [18].

Factors Determining Location

There are several factors to consider before choosing a location. One factor is proximity to a materials recovery facility since our raw material will need to be sorted before we purchase it. Next, the plant needs to be near a hydrogen pipeline since hydrogen is essential to our process. Then, there must be a large population to generate the amount of recycled volume. Finally, the incentives and climate of the state will also be considered.

The first factor is related to the location of material recycling facilities in North America. A materials recycling facility, is a kind of specialized plant that is aimed to receive, separate, and prepare recyclable materials for marketing to end-user manufacturers [31]. The *Figure 8* below, is a map of the 75 largest material recycling facilities in North America, mostly in the United States [32]. Cities with a large population have at least one facility. The following states: California, Texas, Pennsylvania, Connecticut, New Hampshire, and Illinois, have more than 5 facilities.

Figure 8: Map of Material Recovery Facilities in the United States [32]

The next factor is the proximity to sources of hydrogen, since hydrogen is essential to the process. According to the Department of Energy, unlike natural gas, the hydrogen plants are located near large hydrogen consumers. For example, many hydrogen plants are near petroleum refineries and chemical plants in the Gulf Coast region [33]. In order to avoid building a new hydrogen plant, our plant may be built near an existing hydrogen plant. Therefore, Texas may be the most suitable place. The Houston Area has many hydrogen plants due to the large demand of hydrogen from the petroleum industry [34].

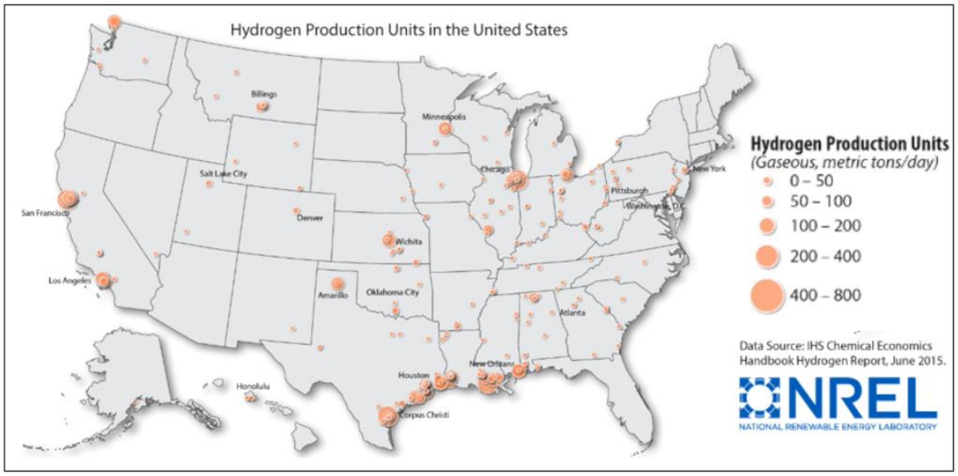


Figure 9: Map of Hydrogen Production in US [34]

The list of locations was narrowed down to three locations based on the proximity of a hydrogen pipeline, large material recovery facility, and incentives. The three locations identified are California, Texas, and Pennsylvania.

The first location considered is Texas, just outside of Houston. Texas has many hydrogen production facilities near Houston due to the nearby petroleum industry [34]. In order to produce gasoline, hydrogen is needed. Having nearby hydrogen plants means that there is easy access to hydrogen, which will be more cost effective for our process. Houston is also a large city with 2.3 million residents [35]. Additionally, there is a material recovery facility that can sort plastic [36]. Overall, the large population and material recovery facility make Houston a good candidate to produce the volume of plastic required for fuel production. Texas also has an oil recovery initiative that can lower the state tax rate. The two incentives that would apply are the Enhanced Oil Recovery (EOR) Incentive and the Enhanced Oil Recovery Using Anthropogenic Carbon Dioxide Incentive [37].

The second location considered is California. Two locations that have hydrogen production plants are Los Angeles and San Francisco. Both cities have large populations. Los Angeles has a population of 3.9 million and San Francisco has a population of 883,000 [35]. Material recovery facilities are also located in both cities [36]. However, California in general is not an ideal location because the state has set a goal to reduce the overall amount of petroleum used by fifty percent by 2030 [38]. In order to meet this goal, California is investing in high-speed rail and pushing for higher fuel economy for cars and trucks. The initiative would decrease the need for petroleum-based fuel in the upcoming years

The third location considered is Pennsylvania. Pennsylvania does have a few hydrogen plants near Pittsburg. However, these plants are small capacity and only produce 50 metric tons or less per day [34]. Pittsburg does have a materials recovery facility that could process and sort plastic, which means that Pittsburg has the sorting techniques that are desired [36]. The population of Pittsburg is smaller at 301 thousand people [35]. Due to the small population of Pittsburg, The State of Pennsylvania does offer tax incentives. The most applicable tax incentives include the Keystone Opportunity Incentive, Research and Development Incentive, and Tax Credit for the creation of Jobs. However, Pittsburg is not an optimal location due to the small population [39].

In conclusion, Houston, Texas is the most optimal location for our plant due to the proximity of hydrogen production facilities and material recovery facilities, large population, and incentives.

Facility Operations

To determine the number of batches the facility will run per year, data regarding the amount of HDPE plastic produced in the United States. According to the Environmental Protection Agency's Advancing Sustainable Materials Management 2017 Report, the amount of durable HDPE recycled in 2016 was 1,470,000 tons [40]. The US population reported in July 2019 is 328 million people [41]. Next, a ratio can be used to determine the amount of plastic produced by the city of Houston. For the design of the plant, the collection area for plastic includes Houston, Dallas, and the Fort-Worth area. The combined population is 4.490 million people.

$$\frac{1{,}470{,}000\ tons\ of\ \ HDPE\ recycled\ in\ US\ per\ year}{328{,}000{,}000\ people\ in\ US} = \frac{x\ tons\ of\ \ HDPE\ recycled\ in\ H.D.FW Areas}{4{,}490{,}000\ people\ in\ H.D.FW\ \ Areas}$$

Solving, we find that the Houston/Dallas/Ft. Worth area can produce 20,122 tons of HDPE per year. Converting to metric tons: 18, 254 metric tons of HDPE per year. The plant will process 50 metric tons of HDPE per day. Using data from Polycylc, a pilot plant in India, we estimate that this process will produce 5.8 barrels of fuel per ton of plastic [5, pp.172].

Appendix 2: Block Flow Diagram

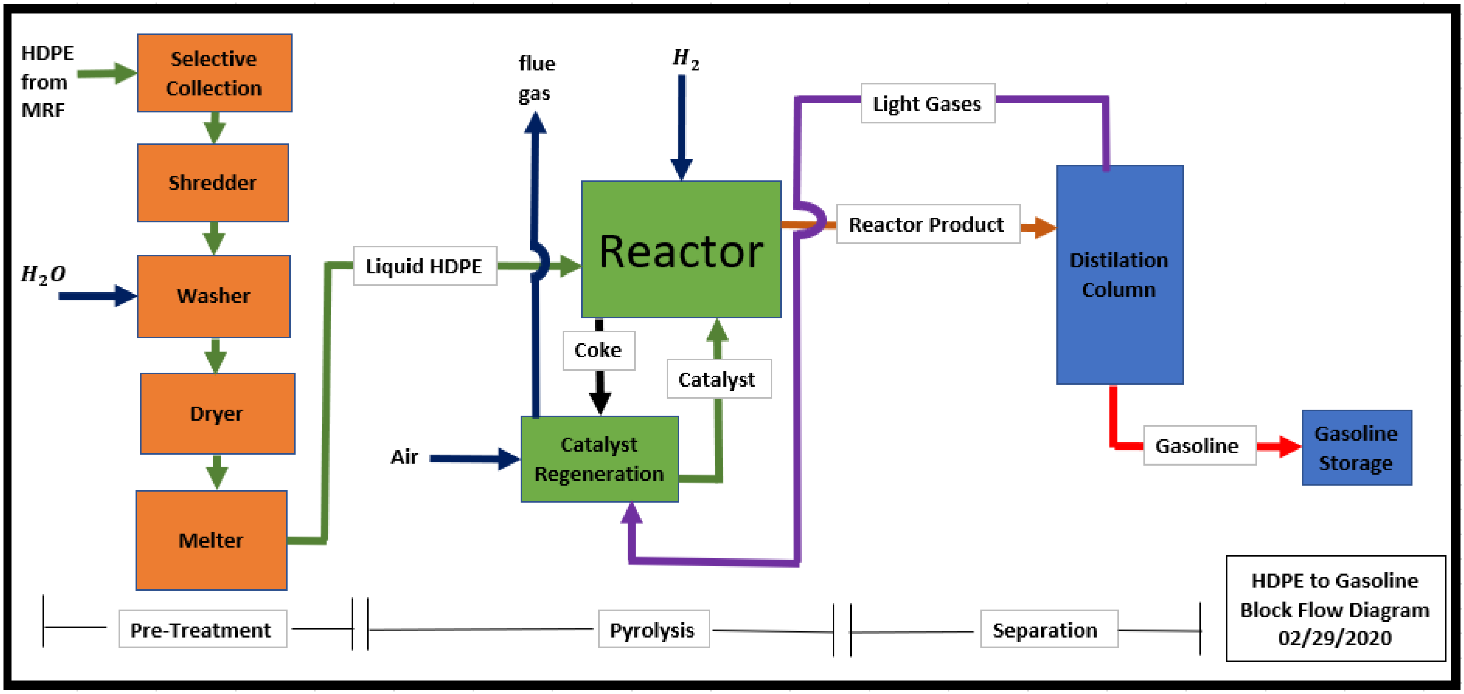


Figure 6: *Block Flow Diagram*

Appendix 3: Full PFD

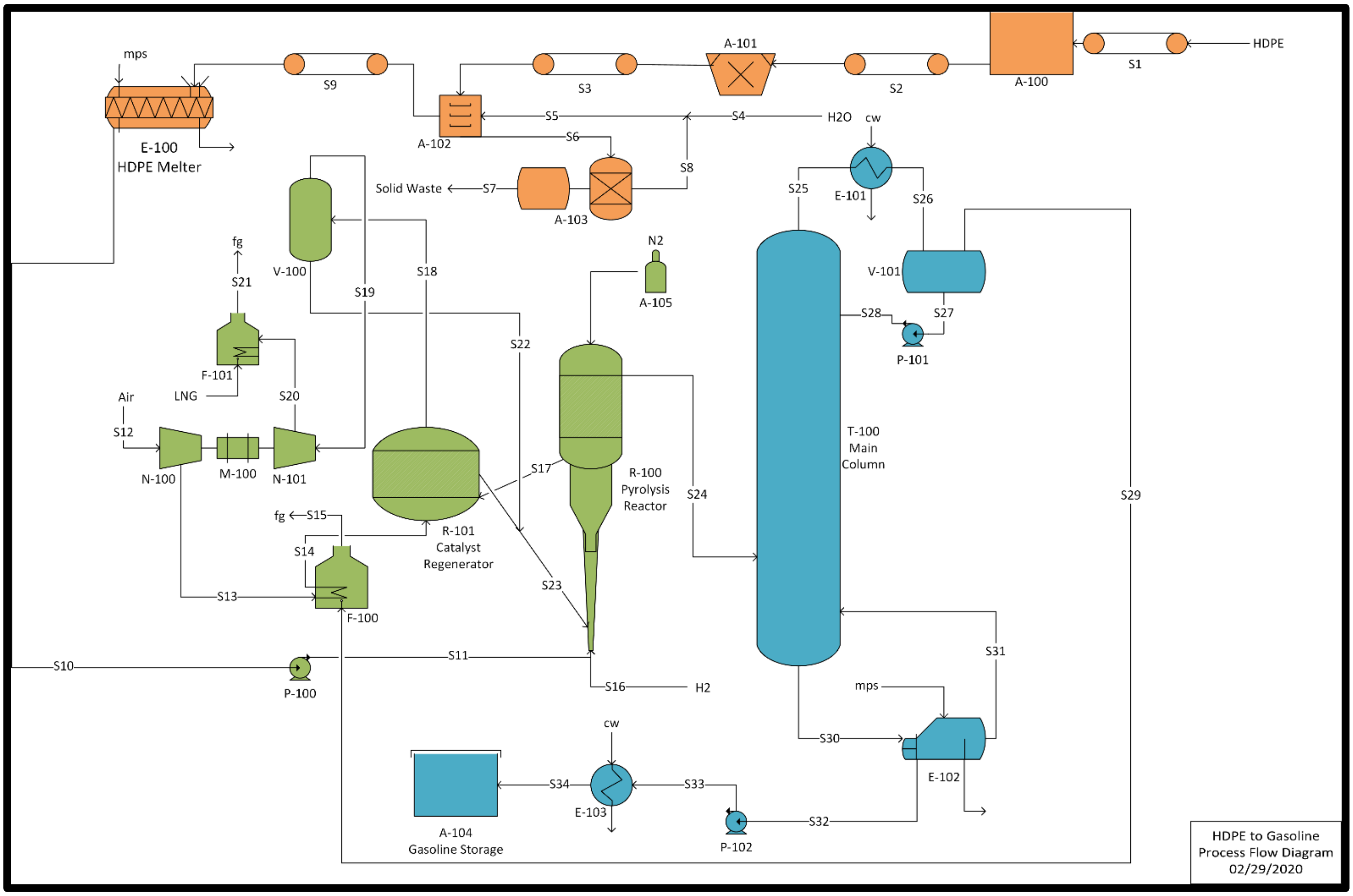


Figure 10: Process Flow Diagram

Appendix 4: Full Mass Balance

Table 2: Pre-Treatment Mass Balance

Pre-Treatment Stream Data										
Stream	1	2	3	4	5	6	7	8	9	10
FROM	HDPE Feed	A-100	A-101	Reservoir		A-102	A-103	A-103	A-102	E-100
TO	A-100	A-101	A-102		A-102	A-103	Treatment		E-100	P-100
Phase	Solid	Solid	Solid	Liquid	Liquid	Liquid	Solid	Liquid	Solid	Liquid
Temperature (°C)	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	120.0
Pressure (bar)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Mass Balance (lb/hr)										
HDPE	4593.3	4593.3	4593.3						4593.3	4593.3
Solid Contaminants							Waste			
H_2O				237.3	4745.7	4508.4		4508.4		
Total Flow	4593.3	4593.3	4593.3	237.3	4745.7	4508.4	0.0	4508.4	4593.3	4593.3

Assumption of Pre-Treatment Process

All coloring, additives, and paper brought in with the HDPE feed from municipal waste treatment facilities are removed during grinding, washing, and drying process. These components are what construct the solid waste gathered from the washing water. The washing water is sent to a water treatment facility to reuse most of the washing water in the continuous process. The recycled washing water is assumed to have a 95 wt% recovery. Thus, only 5 wt% needs to be resupplied. The washer/dryer unit is assumed to be 100% efficient in removing all impurities and fully drying the HDPE feed to the HDPE Melter. The HDPE Melter is a screw conveyor where the HDPE feed is melted be medium pressure steam.

Table 3: Pyrolysis Mass Balance

Pyrolysis Stream Data									
Stream	11	12	13	14	15	16	17	18	19
FROM	P-100		N-100	F-100	F-100	Pipeline	R-100	R-101	V-100
TO	R-100	N-100	F-100	R-101		R-100	R-101	V-100	N-101
Phase	Liquid	Gas	Gas	Gas	Gas	Gas	Solid	Gas/Solid	Gas
Temperature (°C)	123.4	25.0	300.0	482.2	482.2	25.0	450.0	482.2	482.2
Pressure (bar)	3.6	1.0	1.0	1.0	1.0	1.0	3.6	3.6	3.6
Mass Balance (lb/hr)									
HDPE	4593.3								
H_2						0.1			
MCM-41							45.8		
Coke									
CO								335.0	335.0
CO_2					4.23E-03			83.3	83.3
Air		0.8	0.8	0.8					
Total Flow	4593.3	0.8	0.8	0.8	4.23E-03	0.1	45.8	418.3	418.3

Table 4: Pyrolysis/Separation Mass Balance

Pyrolysis/Separation Stream Data									
Stream	20	21	22	23	24	29	32	33	34
FROM	N-101	F-101	V-100	R-101	R-100	V-101	E-102	P-102	E-103
TO	F-101			R-100	T-100	F-100	P-102	E-103	A-104
Phase	Gas	Gas	Solid	Solid	Gas	Gas	Liquid	Liquid	Liquid
Temperature (°C)	25.0	30.0	482.0	482.2	450.0	25.7	102.6	102.7	30.0
Pressure (bar)	1.0	1.0	1.0	1.0	3.6	3.3	3.3	4.1	1.0
Mass Balance (lb/hr)									
C_3H_6					4.6	4.6	2.7E-06	2.7E-06	2.7E-06
C_3H_8					303.2	303.3	5.7E-04	5.7E-04	5.7E-04
C_5H_{12}					992.1	7.6	984.5	984.5	984.5
C_6H_{14}					891.1	6.2E-03	891.1	891.1	891.1
C_7H_{14}					128.6	2.3E-05	128.6	128.6	128.6
C_8H_{18}					119.4	1.3E-07	119.4	119.4	119.4
C_4H_{10}					38.1	38.0	0.1	0.1	0.1
C_4H_8					704.9	699.4	5.5	5.5	5.5
C_5H_{10}					13.8	0.4	13.4	13.4	13.4
C_6H_{12}					78.1	1.3E-03	78.1	78.1	78.1
C_8H_{16}					82.6	1.8E-07	82.7	82.7	82.7
C_9H_{20}					18.4	2.7E-10	18.4	18.4	18.4
$C_{10}H_{22}$					280.1	5.7E-11	280.2	280.2	280.2
C_7H_{14}					487.1	1.5E-05	486.1	486.1	486.1
CO	335.0								
CO_2	83.3	418.3							
MCM-41			Negligable	45.8					
Total Flow	418.3	418.3	0.0	45.8	4141.9	1053.3	3088.0	3088.0	3088.0

<u>Assumptions of Catalyst Regeneration/Pyrolysis</u>

Assumption of Flue Gas and Combustion

Complete combustion is assumed in F-100 for heating up the air into the catalyst regenerator. The mass of carbon dioxide released is determined from the energy needed for F-100. Flue gas exiting the catalyst regenerator is assumed to be 80 wt% carbon monoxide and 20 wt% carbon dioxide from the incomplete combustion of coke on the MCM-41 catalysts. The flue gas exiting the catalyst separator is assumed to be the same temperature exiting the catalyst regenerator. The flue gas is then sent to an expander turbine to convert the heat supplied into energy for the motor to power the air compressor. The flue gas leaving the expander is assumed to be room temperature for maximum energy conversion. The flue gas then enters a fired heater, F-101, where all carbon monoxide is assumed to be combusted into carbon dioxide. Heat of formation was used to calculate the exiting temperature of the flue gas from F-101.

Assumption of Catalyst Regeneration

Few catalyst particles are assumed to exit the catalyst regenerator with the flue gas. To maximize catalyst usage and lower costs, the flue gas enters a catalyst separator to save the few catalyst particles. These particles then join the flow of catalysts returning into the pyrolysis reactor. 100% catalyst regeneration is assumed.

Assumption of Nitrogen Required

Nitrogen is needed for an inert atmosphere within the pyrolysis reactor to avoid autoignition. The nitrogen is supplied by compressed cylinders delivered by monthly supply. The reactor is purged with nitrogen before the plant is started up. A continuous supply of nitrogen is needed. Material flowing in and out of the reactor displaces the nitrogen capacity in the reactor. Thus, the nitrogen supplied to the reactor fluctuates upon flow to keep a fully inert atmosphere. Nitrogen is considered a utility, so it is not presented in the mass balances.

Assumption of Hydrogen Required

To calculate the amount of hydrogen required, one must understand the depolymerization reaction within the pyrolysis reactor. HDPE is a long chain of monomers that are broken into smaller carbon chains. However, it is unknown as to where these breaks concur. Thus, how many and what compound formed from each HDPE molecule is unknown. Understanding how these breaks happen is essential, as it determines the amount of hydrogen needed. Hydrogen is needed as a hydrogen atom needs to be added to newly formed molecules at every break. From this knowledge, it can be assumed that one hydrogen molecule is needed for each break in the HDPE molecule of the depolymerization reaction. Since every break formed two new molecules, it can be understood that the number of moles leaving the reactor is equal to how many moles of hydrogen is needed. However, one must also consider the structure of the HDPE molecules. At the end of the HDPE molecules, there are two CH3 compounds. Thus, hydrogen is not needed at those points, leaving one less hydrogen needed per HDPE molecule. As a result, the moles of hydrogen are the difference of moles between the products and the HDPE feed to the pyrolysis reactor as shown below.

$$M_{HDPE} - M_{Products} = M_{H_2}$$

Appendix 5: Sample Calculations

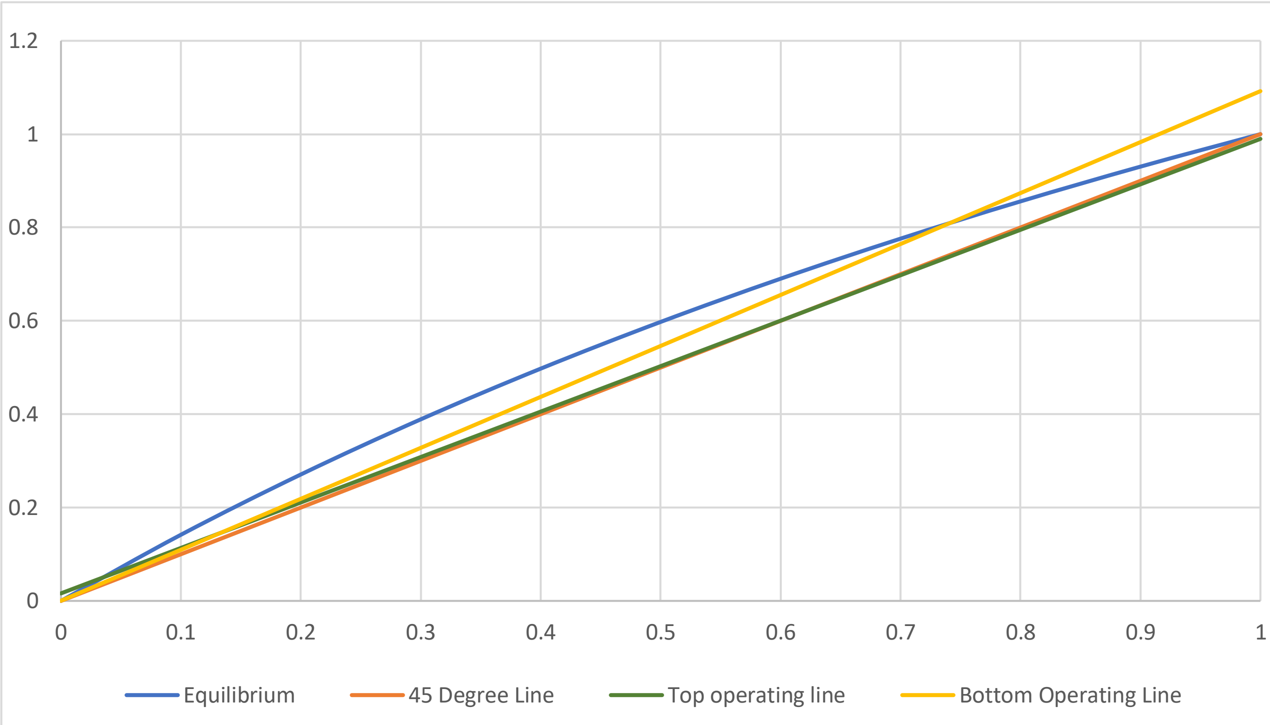


Figure 11: McCabe Thiele Diagram for T-100 Distillation column

In order to plot the equilibrium line of the McCabe Thiele plot the following equation was used:

$$Y = \frac{\alpha x}{1 + (\alpha - 1)x}$$

α = geometric average relative volatility
The α_{avg} can be calculated by using the following equation:

$$\alpha_{avg} = \sqrt{\alpha_{LK} * \alpha_{HK}}$$

α_{LK} = relative volatility of the light key component
α_{HK} = relative volatility of the heavy key component

$$\alpha_{avg} = \sqrt{2.4 * .92} = 1.48$$

This can then be used in order to obtain the equation for the equilibrium curve:

$$Y = \frac{1.48X}{1 + .48X}$$

The equation used for the top operating line is:

$$Y_D = \frac{L}{L+D}X + \frac{D}{L+D}X_D$$

L = The liquid flow rate that is being returned to the column in the top of the column
D = The flow rate of the distillate

The equation used for the bottom operating line was:

$$Y_B = \frac{V'+B}{V'}X + \frac{B}{V'}X_B$$

V' = The boil up rate
B = Liquid flow rate at the bottom of the distillation column
These equations listed can then be used in order to obtain the McCabe Thiele diagram that is shown above in Figure 11. The stages can then be drawn.

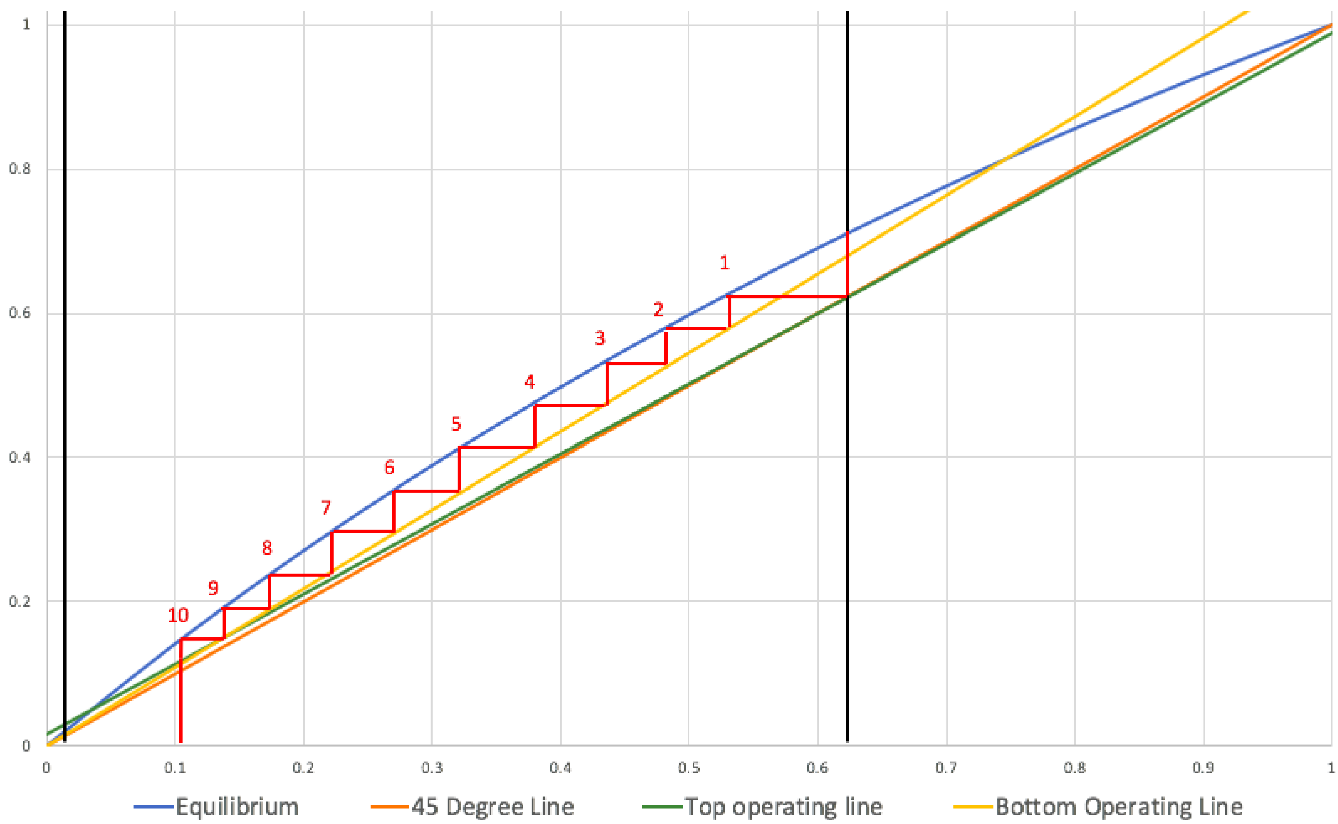


Figure 12: McCabe Thiele Diagram with Stages

It can be seen in the McCabe Thiele diagram that the composition of the light key in the distillate is 61%. This is where the stages can be started. From the graph it can be estimated that there are 10 stages in the distillation column. To confirm this the Fenske equation can be used:

$$N_{min} = \frac{\log\left(\frac{X_{LK}}{X_{HK}}\right)_d \log\left(\frac{X_{HK}}{X_{LK}}\right)_b}{\log(\alpha)}$$

X_{LK} = Moles of the light key component in the distillate
X_{HK} = Moles of the heavy key component in the distillate

$$N_{min} = \frac{\log\left(\frac{39.67\ mol}{1\ mol}\right)_d \log\left(\frac{53.89\ mol}{1\ mol}\right)_b}{\log(2)} = 9.19$$

This can then be rounded up to 10 because there are no such things as partial stages. This agrees with the McCabe Thiele Diagram.

<u>Height Calculation</u>

Aspen Plus estimates that the spacing between the plates is 0.6 meters, and hand calculations show that there are an estimated 10 plates within the column. Using the known information, the height of the column can be calculated by multiplying the plates and the spacing between them.

10 plates * .6 meters per plate = 6 meters

<u>Diameter Calculation</u>

$$u_v = (-.171L_t^2 + .27L_t - .047)\left[\frac{\rho_L - \rho_v}{\rho_v}\right]^{\frac{1}{2}} = .46005$$

u_v = the maximum allowed vapor velocity (m/s)
L_t = Plate spacing (m)
ρ_L = Density of liquid product
ρ_v = Density of vapor product

$$u_v = (-.171(.6\ m)^2 + .27(.6\ m) - .047)\left[\frac{.5806\frac{g}{cc} - .00773\ \frac{g}{cc}}{.00773\ \frac{g}{cc}}\right]^{\frac{1}{2}} = .46005\frac{m}{s}$$

$$D_c = \sqrt{\frac{4V_w}{\pi\rho_v u_v}}$$

D_c = Column Diameter
V_w = maximum vapor rate (kg/s)

$$D_c = \sqrt{\frac{4(.5260412\frac{kg}{s})}{\pi * .7731034\frac{kg}{m^3} * .46005\frac{m}{s}}} = 1.37\, meters$$

Table 5: Hand Calculation Results

Number of Plates	10
Diameter	1.37 meters
Height	6 meters

Aspen Simulation

As Aspen Plus v11 simulation was performed for the pyrolysis unit and column. The simulation was focused on this section as Aspen Plus v11 is limited in its capabilities to simulate solids, especially because HDPE would have to be manually defined. In order to avoid any simulation inaccuracies, this selection was chosen. Coke production and the catalyst were omitted from the simulation also to avoid inaccuracies. The simulation utilized a yield reactor for the pyrolysis unit and a RadFrac column for the separation step. The column was optimized by initially using a DISTWU column in order to find the minimum reflux ratio and number of stages. The simulation flow diagram can be seen below in Figure 13.

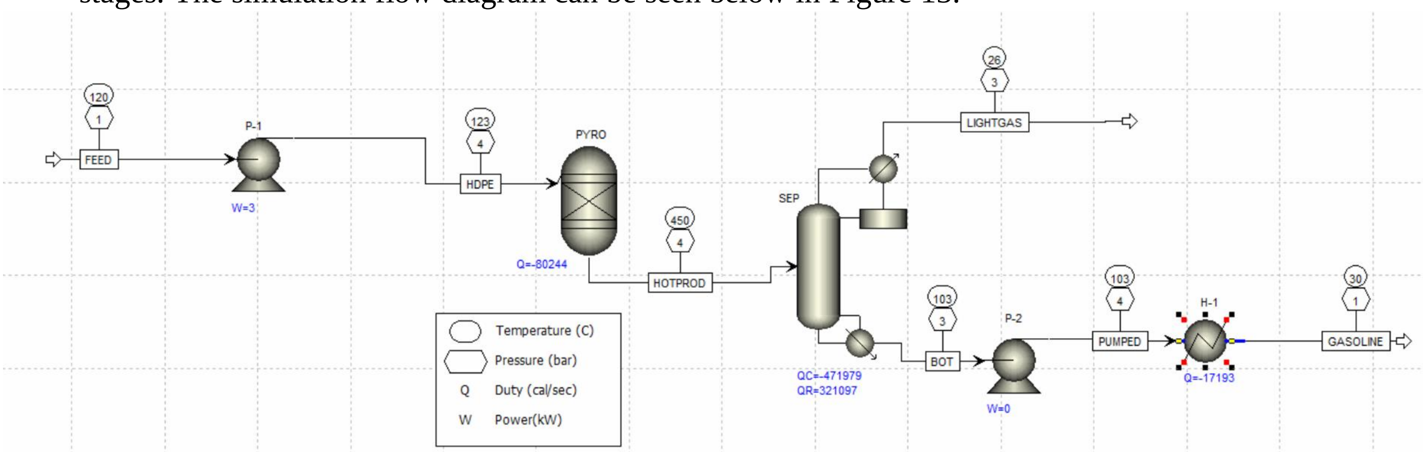


Figure 13: Aspen Flow Diagram

The column separation can be seen in Figure 14 in terms of vapor mole fraction at each stage. As expected, the lighter gases of propane and butane are at their highest mole fractions at the top of the column. All of the heavier hydrocarbons can be seen primarily at the bottom of the column as shown in Figure 14.

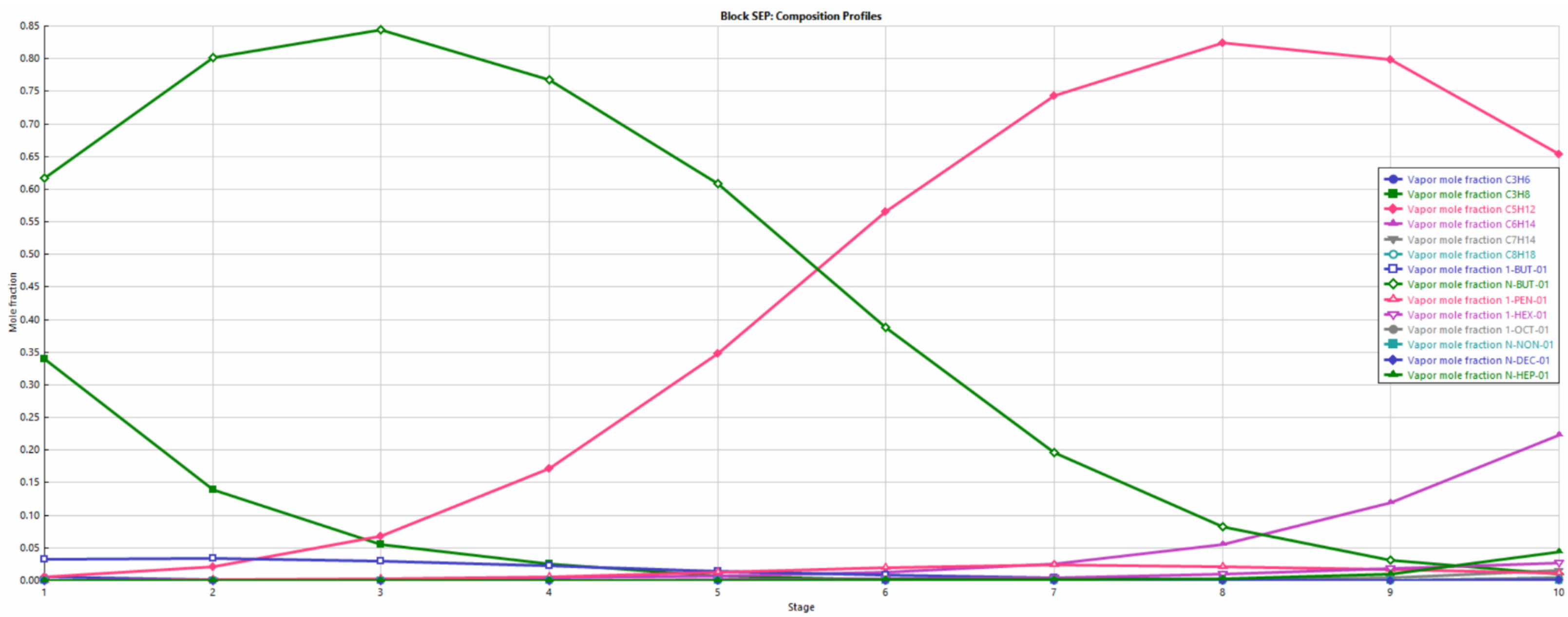


Figure 14: Vapor Mole Fraction Separation at Each Column Stage

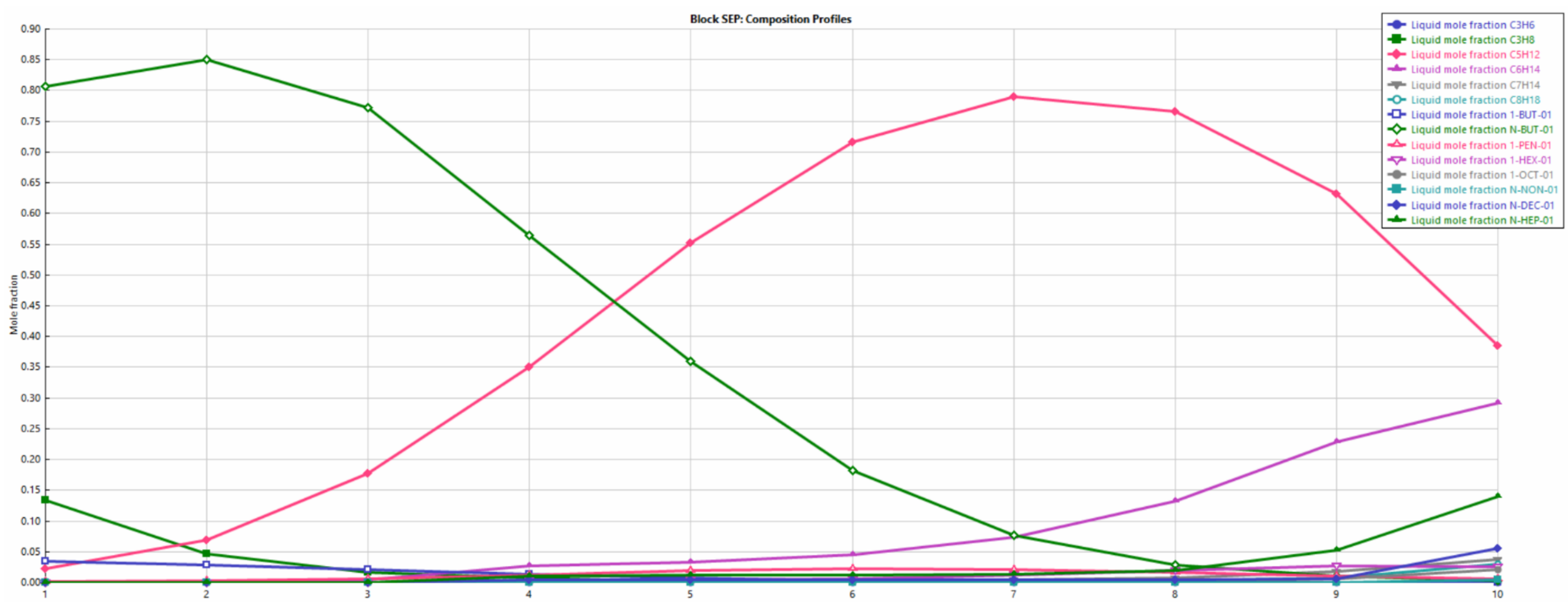


Figure 15: Liquid Mole Fraction Separation at Each Column Stage

Few assumptions were made in the Aspen Plus simulation. The first being that the DISTWU results would completely transfer over to the RadFrac column model, even though not all of the same inputs were required. Another assumption was made that the system could run at a pressure of about 50 psi. This was done to in order to aid the separation and raise the temperature of the column's top product stream. If this was not done, the stream would have been too cold, requiring either a very large column or a large load of a colder utility.

Calculation Comparison

Table 6: Column Result Comparison

	Hand Calculation Results	Aspen Results
Column Trays	9.1 Trays	9.8 Trays
Column Diameter	1.3 m	1.2 m
Column Height	5.4 m	5.0 m

Table 6 compares the features of the separation column that were hand calculated versus the Aspen Plus simulation results. The column trays from both sources were found to be only slightly different, in which rounding up to the next whole tray number would equate the same amount of trays. This occurred as both numbers are generated by the Fenske equation; however due to Aspen's greater capability to calculate component interaction, the Aspen value was slightly different. As for diameter, the difference in calculation was also minimal. It can be assumed that the same behind the scenes calculations were done in Aspen as were shown in the hand calculation. Finally, the overall column height poses the greatest difference. In the hand calculations, it was assumed that all of the column height would come from spaces between trays whereas Aspen may have optimized that spacing even further. Aspen most likely assumed a smaller value than 0.6 meters for spacing in between the trays.

Appendix 6: Equipment List and Sizing

Table 7: Equipment Sizing

Equipment	Label	Material	Sizing
HDPE Feed Conveyor	S1	Carbon Steel with Rubber Belt	Open belt conveyor, Length = 100 feet, Belt width = 42 in, Belt Capacity = 4593.3 lb/hr
HDPE Storage	A-100	Carbon Steel	Diameter = 5 feet, Height = 8 Feet (Storage = 1,500 Gallons)
HDPE Feed Conveyor	S2	Carbon Steel with Rubber Belt	Open belt conveyor, Length = 100 feet, Belt width = 42 in, Belt Capacity = 4593.3 lb/hr
Grinder	A-101	Carbon Steel	Wight = 9800 lbs, Crusher flow rate = 6000 lb/hr
Grinded HDPE Conveyor	S3	Carbon Steel with Rubber Belt	Enclosed belt conveyor, Length = 100 feet, Belt width = 42 in, Belt Capacity = 4593.3 lb/hr
Washer/Dryer	A-102	Carbon Steel	Diameter = 7 feet, Height = 8.5 Feet (Volume = 3,000 Gallons)
Washed HDPE Conveyor	S9	Carbon Steel with Rubber Belt	Enclosed belt conveyor , Length = 100 feet, Belt width = 42 in, Belt Capacity = 4593.3 lb/hr
HDPE Melter (Screw Conveyor)	E-100	Carbon Steel with interior firebrick linings	Diameter = 14 in @ 30 % Trough Loading, Max Capacity = 1,040 ft^3/hr
Centrifugal Pump	P-100	Carbon Steel	1.5 HP
Pyrolysis Reactor	R-100	Carbon Steel with interior firebrick linings	Diameter = 4 feet, Height = 8.5 Feet (Volume = 800 Gallons)
Compressor	N-100	Carbon Steel	Weight = 9900 lbs, 75 HP
Air Furnace	F-100	Carbon Steel with interior firebrick linings	Weight = 4700 lbs, 17000 BTU/h
CO Boiler	F-101	Carbon Steel with interior firebrick linings	Weight = 6600 lbs, 182000 BTU/h
Catalyst Separator	V-100	Carbon Steel	Diameter = 3.5 feet, Height = 5.5 Feet (Volume = 400 Gallons)
Expander	N-101	Carbon Steel	10kw, 13.6 HP
Catalyst Regenerator	R-101	Carbon Steel with interior firebrick linings	Diameter = 5 feet, Height = 6.5 feet (Volume = 130 ft^3)
Trayed Distillation Tower	T-100	Carbon Steel	Height = 33 feet , 10 Trays
Floated Head Heat Exchanger	E-101	Carbon Steel	55 Tubes in 1 Shell, Tube Length 12 Feet, Tube passes = 1

Reflux Drum	V-101	Carbon Steel	1000 gallons (half drums capacity full and 5 minute holdup)
Centrifugal Pump	P-101	Carbon Steel	10.0 HP
Reboiler	E-102	Carbon Steel	18 tubes in 1 shell, Tube Length 8 Feet, Tube passes = 2
Centrifugal Pump	P-102	Carbon Steel	3.0 HP
Shell in Tube heat Exchanger	E-103	Carbon Steel	7 Tubes in 1 Shell, Tube Length 8 Feet, Tube passes = 2
Horizontal Drum	A-104	Carbon Steel	Diameter = 24 feet Height = 32 Feet (Storage = 105,000 Gallons)

Conveyor Assumptions:

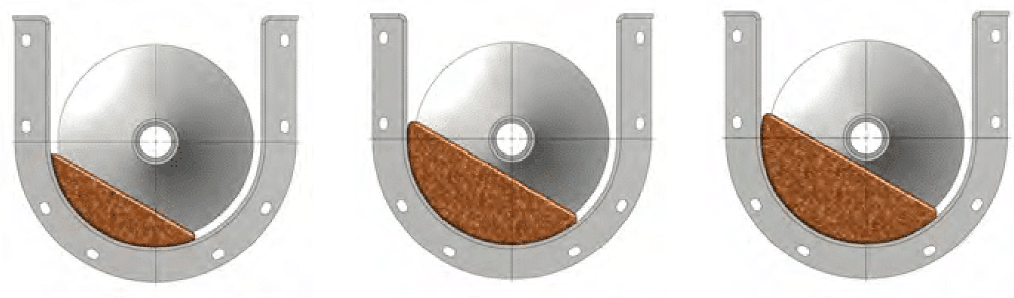
The raw material of HDPE comes from MRF, and it needs to be transported by conveyor belt. The size of these conveyor belts is based on some assumptions. The first one is related to the length. In the reality, the actual length depends on the space arrangement of other equipment, and the length of conveyor can be any suitable length. In this project, a length of 100 feet was chosen because this value is the most common. [42] The second one is the width of conveyor. A 42-inch width was selected since it can convey a large variety of sizes of HDPE plastics.

Storage and Washer/Dryer Assumptions:

For this equipment, the appearance of disturbances like suddenly increasing flow rate of the feed stream, or any other sudden situations when designing the size of them need to be considered. For example, the normal input and output stream of the storage unit is 4593.3 lb/hr, nearly 600 gallon/hr. In order to have enough space to store the HDPE, the volume must be larger than the previous one. Thus, we choose to have 1500-gallon storage is chosen. The same reason is applied to the washer/dryer unit as well. Designing a larger volume is needed in case of disturbances.

Screw Conveyor Assumptions:

In order to size the Screw Conveyor there were a few assumptions that had to be made. First, the trough loading needed to be estimated. There were options of 15%, 30% and 45% as shown in the diagrams below



15 % Loading 30 % Loading 45 % Loading

Figure 16: Screw Conveyor Loading

Using different trough loadings will allow for different amount of solid to be transported. This will also change the contact that the particles have with the wall. This will change the length of the screw conveyor. A 30 % loading was chosen because it experienced the best efficiency as shown in the sizing table [43].

Now that a loading size has been chosen, a diameter size is needed. The inlet flow rate of HDPE is 726 ft^3/hr. Following the referenced capacity table [43], it was decided that a diameter of 14 inches. at a rotation speed of 50 RPM. This has a max capacity of 1,040 ft^3/hr. This gives room for error but is also above the 12 in, 50 RPM capacity of 645 ft^3/min.

Appendix 7: Utilities

Table 8: Utilities Usage

Description	Type	Units	Amount
Cooling Water	**Water**	**lb/hr**	**768,000**
	Condenser	*lb/hr*	*763,000*
	HX-E103	*lb/hr*	*27,700*
Medium Pressure Steam	**Water**	**lb/hr**	**5,240**
	Reboiler	*lb/hr*	*5,240*
	Heated Screw Conveyor	*lb/hr*	*3*
Electricity	**Power**	**kW**	**12**
	Pump, P-100	*kW*	*2.76*
	Pump, P-101	*kW*	*7.5*
	Pump, P-102	*kW*	*0.16*
	Conveyor Motor (4x)	*kW*	*1.5*
Liquid Natural Gas	**Fuel**	*BTU/hr*	**4,163,000**
	Furnace, F-100	*BTU/hr*	*63*
	Furnace, F-101	*BTU/hr*	*4,163,000*
Nitrogen	*Utility*	*lb/hr*	**8**

Table 8 shows the utility usage for this process. Utilities were calculated using ASPEN PLUS or using hand calculations. The units that are using cooling water are the distillation column's condenser and the heat exchanger, E103. The units that are using medium pressure steam are the reboiler and the screw conveyor that melts solid HDPE. In order to calculate the enthalpy of solid HDPE, it was assumed that internal energy is the same as enthalpy. Then, the internal energy at 25°C was used to calculate the amount of steam. [44] Nitrogen is used to create an inert environment in the reactor. The electricity includes the power required for three centrifugal pumps and four conveyor belts. It was assumed that each conveyor belt uses 0.37 kW. [45]

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